In Vivo CRISPR-Cas9 Base Editing Achieves Durable Factor VIII Restoration in a Murine Hemophilia A Model

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Abstract

Hemophilia A, caused by loss-of-function mutations in the F8 gene encoding coagulation factor VIII (FVIII), affects approximately 1 in 5,000 male births worldwide. Current prophylactic treatment with recombinant FVIII requires intravenous infusions every 2–3 days at an annual cost exceeding USD 300,000 per patient. Here we report a single-administration in vivo base editing strategy that converts the most prevalent pathogenic missense mutation (F8 c.6115C>T, p.Arg2038Cys) to a synonymous wild-type codon in hepatocytes using adenine base editor 8e (ABE8e) delivered by a liver-tropic lipid nanoparticle (LNP) formulation. In F8-R2038C knock-in mice, a single intravenous injection of ABE8e LNPs at a dose of 0.5 mg/kg achieves 72 ± 8% on-target A-to-G editing in bulk liver tissue at 4 weeks, restoring plasma FVIII activity to 89 ± 11% of normal levels. Tail-clip challenge bleeding time normalises to wild-type values and remains durable at 52 weeks without detectable immune response or off-target indels at the top 50 predicted off-target sites. These data support progression to non-human primate toxicology studies and position single-dose CRISPR base editing as a curative alternative to life-long protein replacement therapy.

Keywords: CRISPR-Cas9; base editing; hemophilia A; factor VIII; lipid nanoparticle; gene therapy; curative

1. Introduction

Hemophilia A is an X-linked bleeding disorder caused by deficiency of coagulation factor VIII. Over 400 distinct F8 mutations are catalogued, with the p.Arg2038Cys missense variant accounting for 8–12% of severe cases globally. Current standard-of-care consists of prophylactic recombinant FVIII infusion administered two to three times per week to maintain trough FVIII activity above 1–3 IU/dL. Despite improving clinical outcomes, this regimen carries high treatment burden, risk of inhibitor formation in 25–30% of previously untreated patients, and annual treatment costs that place curative therapies out of reach for patients in lower-income settings.

Gene therapy approaches using adeno-associated virus (AAV) vectors have demonstrated multi-year FVIII restoration in Phase 3 trials (HAVEN 6, BioMarin Pharmaceutical). However, AAV-based therapies are limited by pre-existing capsid immunity in 30–50% of adults, liver toxicity at high vector doses, and loss of episomal DNA as hepatocytes divide, with clinical FVIII levels declining 10–15% annually. CRISPR-mediated genomic integration or base correction offers a permanent, cell-division-stable fix to these limitations.

Adenine base editors (ABEs), which catalyse A-to-G transitions without creating double-strand DNA breaks, achieve low indel rates (<0.1%) and are thus particularly suited to precision correction of single-nucleotide pathogenic variants. LNP delivery of mRNA encoding ABE8e avoids immunogenicity concerns associated with viral vectors and enables multiple dosing if needed. Here we engineer a hepatocyte-targeted LNP formulation co-encapsulating ABE8e mRNA and a guide RNA targeting the F8 Arg2038Cys locus, and demonstrate durable haemostasis correction in a faithful knock-in mouse model.

2. Methods

LNPs were formulated using a microfluidic mixer at a molar lipid ratio of ionisable lipid:DOPE:cholesterol:PEG-lipid of 50:10:38.5:1.5. Guide RNA (gRNA) targeting the F8 Arg2038Cys locus (5'-TGGCAATGACCAGTCCTGAG-3') was synthesised with 2'-O-methyl and phosphorothioate end-protective modifications. ABE8e-SpCas9 mRNA (5-methylcytidine + pseudouridine substituted) was produced by in vitro transcription. The gRNA:mRNA mass ratio was optimised at 1:5 to achieve maximal editing efficiency while maintaining LNP particle size below 100 nm and encapsulation efficiency above 95%.

F8-R2038C knock-in mice (C57BL/6 background, n = 12 per group) received a single tail-vein injection at 0.5 mg/kg or 1.0 mg/kg total RNA dose. Editing efficiency was quantified by amplicon deep sequencing of hepatocyte genomic DNA at 4, 12, 26, and 52 weeks. Plasma FVIII activity was measured by one-stage clotting assay. Bleeding time was assessed by the tail-clip haemostasis model. Off-target analysis used GUIDE-Seq at Week 4 in liver tissue across 50 computationally predicted off-target sites.

3. Results

ABE8e LNPs at 0.5 mg/kg achieved 72.3 ± 7.9% A-to-G conversion at the target adenosine in bulk liver at Week 4, determined by amplicon sequencing. FVIII plasma activity recovered to 89.2 ± 11.4% of wild-type at Week 4 and remained stable at 83.7 ± 9.1% at Week 52, compared with 3.1 ± 0.8% in uninjected F8-R2038C mice ($p < 0.0001$, one-way ANOVA). Tail-clip bleeding time normalised to 2.8 ± 0.4 min in treated mice, comparable to wild-type (2.6 ± 0.3 min) and significantly shorter than untreated knockin mice (>10 min, censored).

GUIDE-Seq analysis detected no off-target editing events at ≥0.1% frequency across all 50 predicted off-target sites. Serum ALT and AST levels transiently elevated 2.1-fold at Day 7 but returned to baseline by Day 21, consistent with transient LNP-mediated innate immune activation. No anti-Cas9 T-cell response was detected by IFN-γ ELISpot at any timepoint. Histopathological assessment at Week 52 showed no hepatic fibrosis.

4. Discussion and Commercialisation Pathway

The durability of FVIII restoration at 52 weeks supports the hypothesis that LNP-mediated base editing achieves permanent genomic correction in post-mitotic or slowly dividing hepatocytes. The absence of detectable off-target editing and the favourable hepatotoxicity profile strengthen the case for IND filing. Commercial modelling suggests a single-dose curative therapy priced at USD 2–3 million (in line with approved gene therapies Hemgenix and Roctavian) achieves a favourable incremental cost-effectiveness ratio relative to lifetime prophylaxis above USD 300,000/year. Vertex Pharmaceuticals and Intellia Therapeutics have initiated licensing discussions for the LNP-ABE8e platform.

5. Conclusion

In vivo base editing with ABE8e LNPs provides durable, safe correction of the F8 Arg2038Cys mutation in a murine hemophilia A model. These results establish proof-of-concept for a single-administration curative therapy and define the preclinical data package required for regulatory filing.

References

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